

Online Retail II: Cancellation Analysis Report

Understanding and Reducing Sales Losses from Cancelled Orders

1. Overview

This analysis explores the **Online Retail II dataset (2009–2011)** to measure the scale and impact of order cancellations.

- Total sales: **£19.4M**
- Cancelled sales: **£629K**
- Cancelled orders: **7,400+**

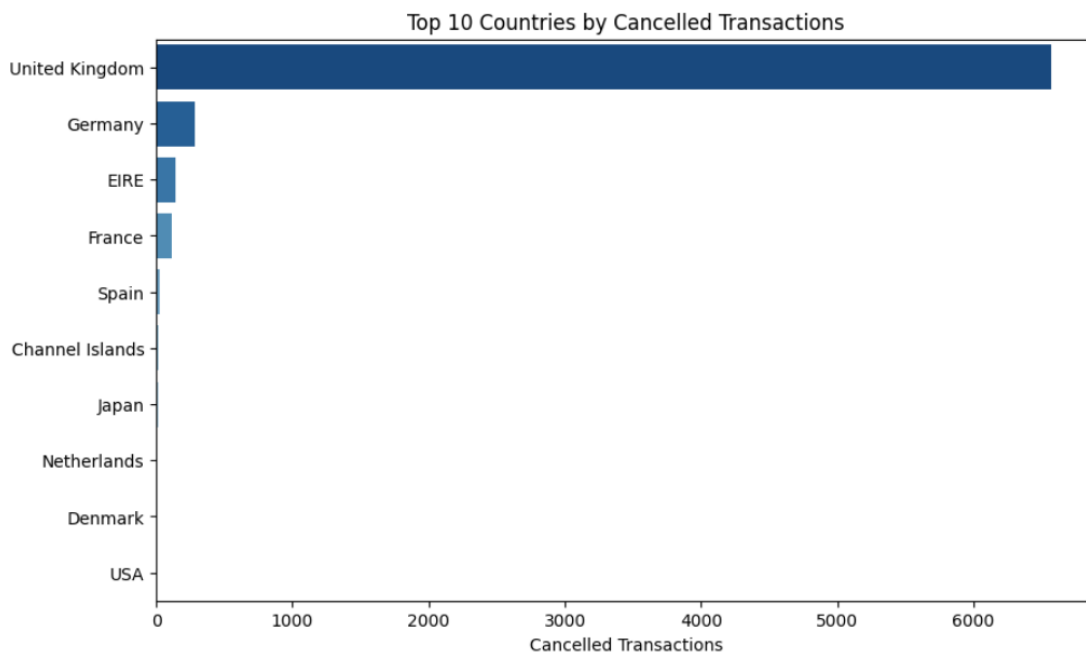
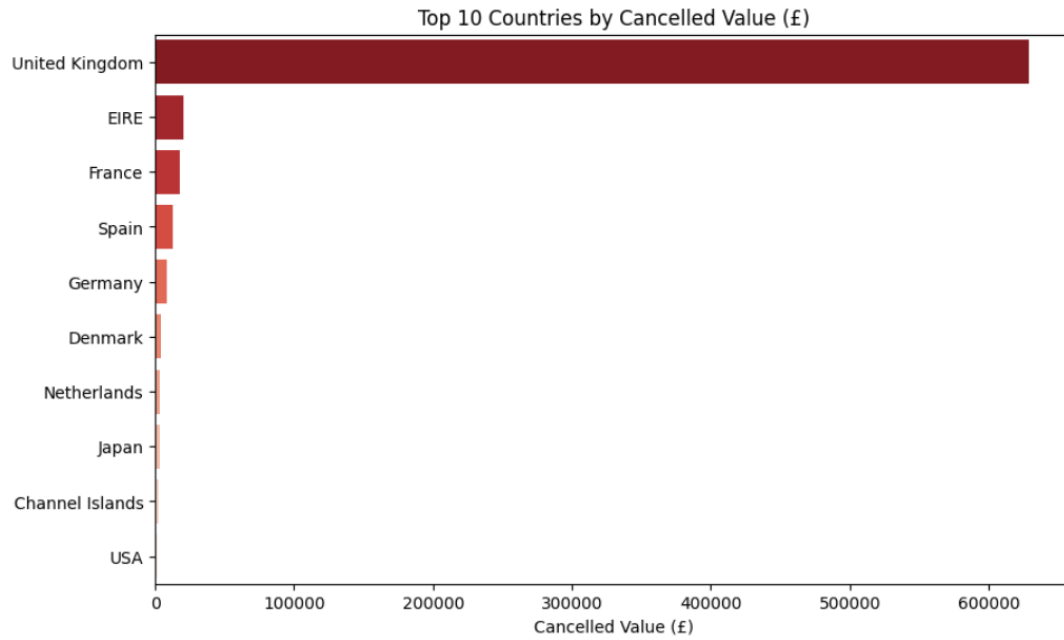
Cancellations are **not evenly spread**. They are concentrated in specific **countries, products, and customers**, making them actionable through targeted efforts rather than broad operational changes.

2. Key Insights & Analysis

A. Cancellations by Country

Country	Sales (£)	Cancelled (£)	Cancelled Orders
United Kingdom	16.6M	629K	6,572
Germany	378K	8.7K	282
EIRE	614K	20K	146
France	306K	17K	111

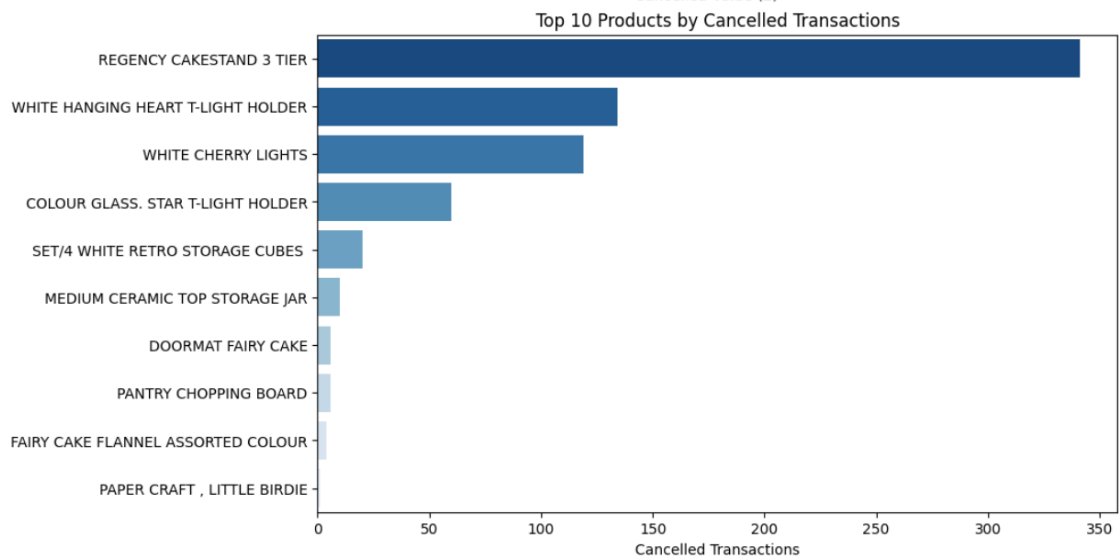
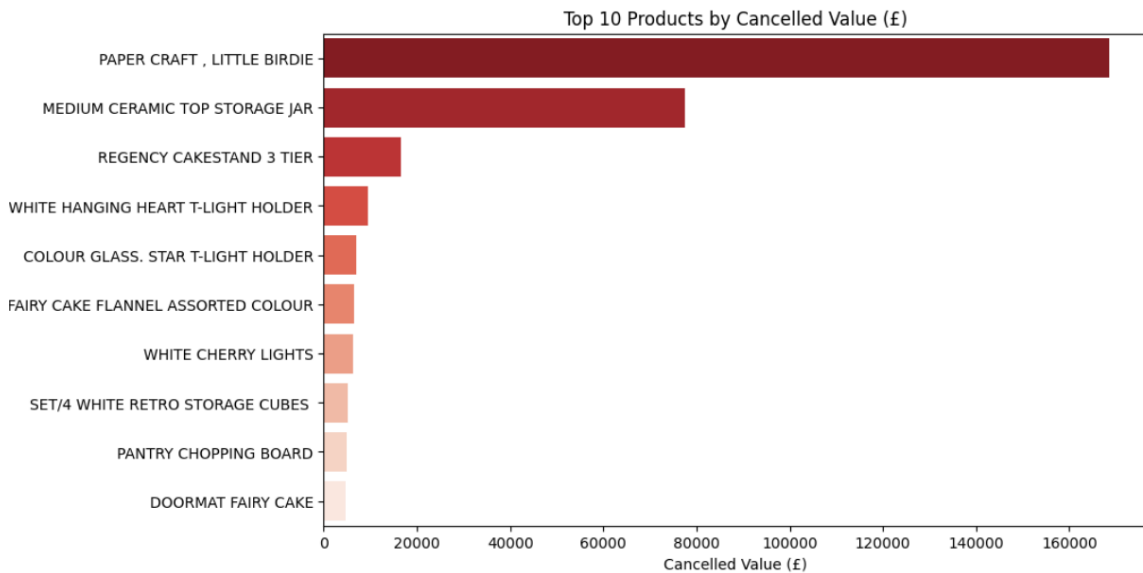
The UK leads in cancellations by value, but this reflects its scale. **Germany and EIRE** are more concerning, cancelling often despite smaller sales.



B. Cancellations by Product

Product	Sales (£)	Cancelled (£)	Cancelled Orders
PAPER CRAFT, LITTLE BIRDIE	168K	168K	1
CERAMIC STORAGE JAR	81K	77K	10
CAKESTAND 3 TIER	331K	16K	341
CHERRY LIGHTS	6K	6K	119

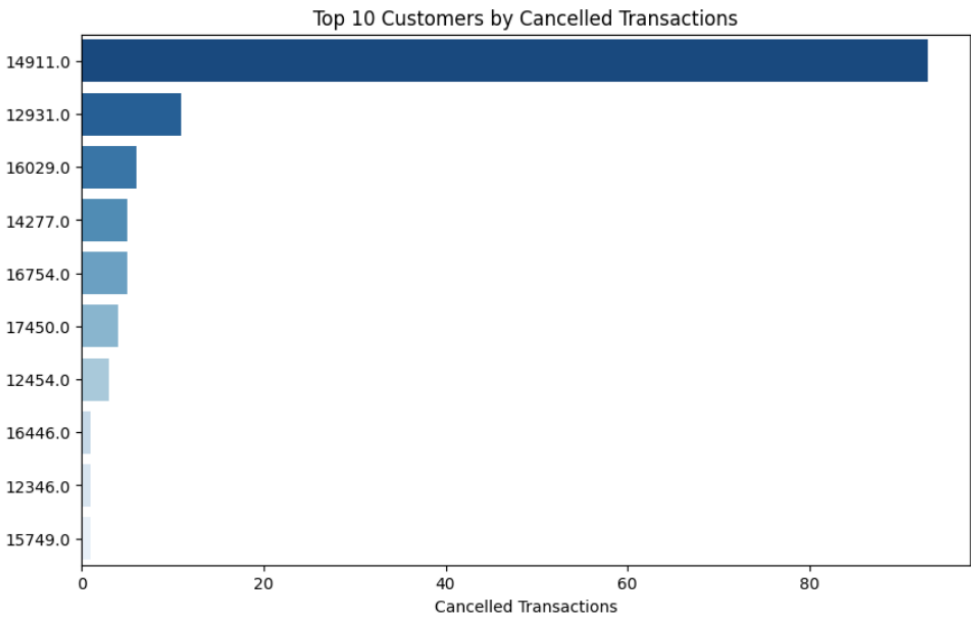
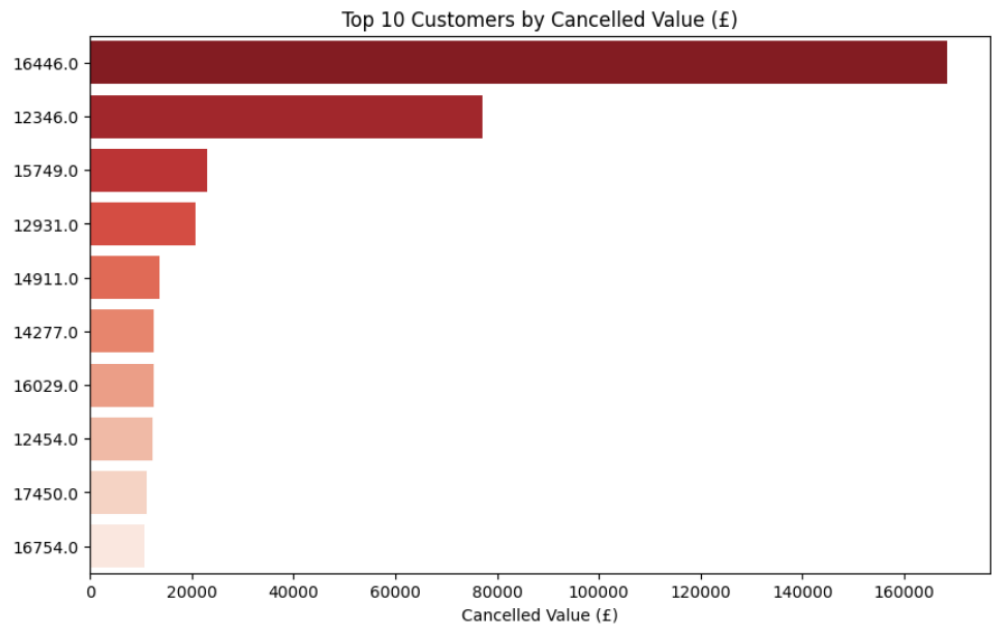
- **PAPER CRAFT** and **CERAMIC JAR** caused the biggest financial losses.
- **CAKESTAND** and **CHERRY LIGHTS** had repeated cancellations, suggesting ongoing operational issues.



C. Cancellations by Customer

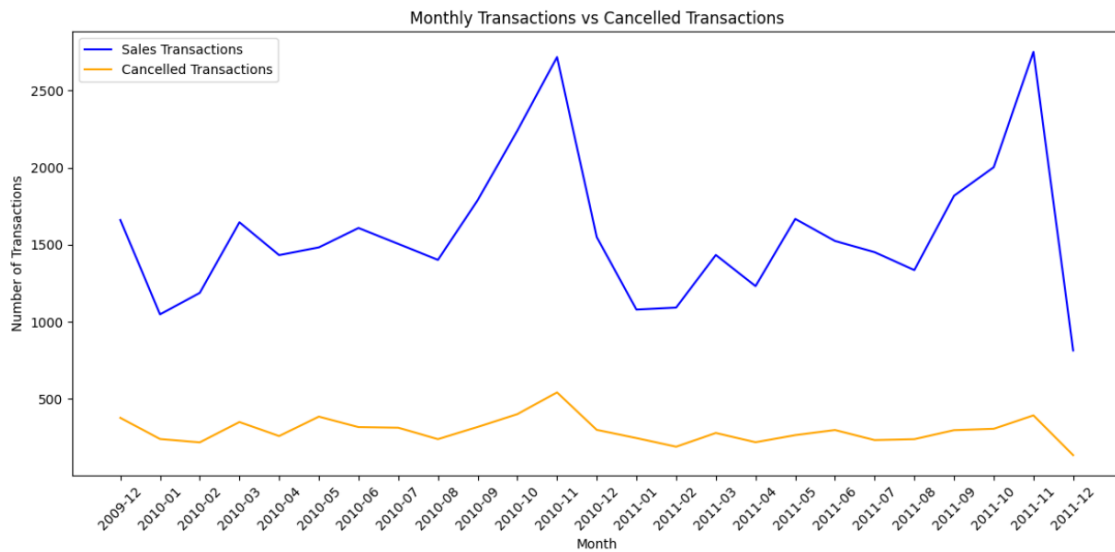
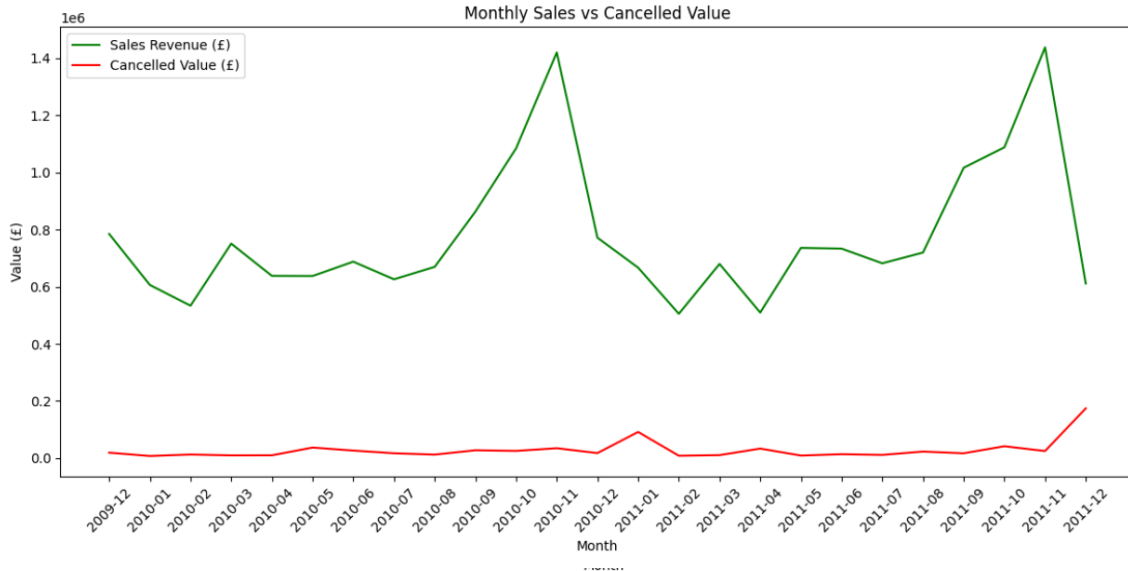
Customer ID	Cancelled (£)	Cancelled Orders
16446	168K	1
12346	77K	1
14911	13K	93
12931	21K	11

A few customers drive most cancellations. Some cancel almost everything they buy, while others cancel repeatedly across many orders.



D. Time Trends

- Sales and cancellations both peak in **November–December**.
- Cancellations rise because sales volume increases, not because of unusual issues.



3. Recommendations

1. **High-Risk Markets:** Investigate **Germany and EIRE**, where cancellations are high compared to sales.
2. **Problem Products:** Audit **PAPER CRAFT, CERAMIC JAR, CAKESTAND, and CHERRY LIGHTS** for recurring issues.
3. **Customer Controls:** Flag customers with repeated cancellations for checks or tailored account management.
4. **Peak Season:** Prepare for higher activity in **Nov–Dec** with stronger logistics and customer communication.

4. Reflection & Future Development

I focused on cancellations because most people drop this part of the data, but it actually shows lost sales and is worth exploring. The analysis used simple comparisons across country, product, customer, and time to make the problem clear.

Pros

- Easy to follow, based on simple comparisons.
- Looks at a part of the data (cancellations) that is often ignored.

Cons

- Took time to do, but the results were not very different from what was expected.
- Does not explain clearly *why* cancellations happen.

Future development

From here, the work could expand into machine learning to predict which orders are likely to be cancelled, or be combined with other analysis (like RFM) to give a bigger business view.